THE FINITE FREE STAM INEQUALITY

Abstract. The classical Stam inequality asserts the superadditivity of the reciprocal Fisher information under convolution of independent random variables. We prove the polynomial analogue in the framework of finite free probability: for monic, degree-n, real-rooted polynomials p and q,

$$\frac{1}{\Phi_n(p \boxplus_n q)} \geq \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)},$$

where $\boxplus_n$ is the symmetric additive convolution of Marcus, Spielman, and Srivastava, and Φ_n is the finite free Fisher information. The proof combines an algebraic inequality—the Score-Gradient Inequality, established via two applications of Cauchy–Schwarz—with a flow-based argument exploiting the Hermite semigroup structure of $\boxplus_n$, together with the Jacobian contraction inequality for the root map. We also derive a closed-form expression for Φ_n via residue calculus and verify the inequality explicitly for cubics.

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1. Introduction

1.1. **Background and motivation.** In information theory, the Stam inequality [5] states that if X and Y are independent random variables with finite Fisher information $I(X)$ and $I(Y)$, then

$$\frac{1}{I(X+Y)} \geq \frac{1}{I(X)} + \frac{1}{I(Y)}.$$

This fundamental inequality—equivalent to the entropy power inequality of Shannon and Stam—captures the principle that convolution of independent sources strictly increases disorder.

Finite free probability, introduced by Marcus, Spielman, and Srivastava [4], provides a polynomial analogue of free probability in which random variables are replaced by real-rooted

Date: February 14, 2026.

polynomials and addition by a deterministic convolution operation $\boxplus_n$. Within this framework, the natural question arises:

Does the Stam inequality hold for the finite free additive convolution?

The purpose of this paper is to answer this question affirmatively.

1.2. **Statement of the main result.** Let $\mathcal{P}_n$ denote the space of monic polynomials of degree n with real coefficients, and $\mathcal{P}_n^{\mathbb{R}} \subset \mathcal{P}_n$ the subset with all real roots. For $p \in \mathcal{P}_n^{\mathbb{R}}$ with distinct roots $\lambda_1 < \cdots < \lambda_n$, define the *scores*

$$V_i := \sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j}$$

and the *finite free Fisher information* $\Phi_n(p) := \sum_{i=1}^n V_i^2$. The symmetric additive convolution $p \boxplus_n q$ is recalled in Section 2.

Theorem 1.1 (Finite Free Stam Inequality)**.** *For $p, q \in \mathcal{P}_n^{\mathbb{R}}$ with positive variance,*

$$\frac{1}{\Phi_n(p \boxplus_n q)} \geq \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)}. \tag{1}$$

The proof combines four ingredients: the Score-Gradient Inequality (Theorem 3.1), a dissipation identity for the Hermite flow (Lemma 4.7), the Score–Jacobian relation (Proposition 5.2), and the Jacobian contraction inequality (Proposition 5.3).

Convention. We set $\Phi_n(p) := \infty$ (equivalently $1/\Phi_n(p) := 0$) when p has a repeated root. The proof below establishes (1) under the hypothesis that p and q have distinct roots; the extension to the boundary of $\mathcal{P}_n^{\mathbb{R}}$ is discussed in Remark 6.3.

2. Preliminaries

2.1. **Root statistics.** For $p(x) = \prod_{i=1}^n (x - \lambda_i) = \sum_{k=0}^n a_k x^{n-k}$ with $a_0 = 1$, the *mean* and *variance* of the root distribution are

$$\bar{\lambda} := \frac{1}{n} \sum_{i=1}^n \lambda_i, \qquad \sigma^2(p) := \frac{1}{n} \sum_{i=1}^n (\lambda_i - \bar{\lambda})^2.$$

Lemma 2.1. $\sigma^2(p) = \dfrac{(n-1)a_1^2}{n^2} - \dfrac{2a_2}{n}$.

Proof. By Vieta's formulas, $\sum_i \lambda_i = -a_1$ and $\sum_{i<j} \lambda_i \lambda_j = a_2$, whence $\sum_i \lambda_i^2 = a_1^2 - 2a_2$. The result follows from $\sigma^2 = \frac{1}{n} \sum_i \lambda_i^2 - \bar{\lambda}^2$. □

2.2. **Symmetric additive convolution.** The *finite free additive convolution* is defined by averaging over the orthogonal group: if A and B are real symmetric matrices with characteristic polynomials p and q, then

$$(p \boxplus_n q)(x) = \int_{O(n)} \det(xI - (A + QBQ^T))\, d\mu(Q).$$

By the MSS theorem [4], this admits a differential operator representation: writing $q(x) = \sum_{k=0}^n b_k x^{n-k}$,

$$(p \boxplus_n q)(x) = T_q\, p(x), \qquad T_q := \sum_{k=0}^n \frac{(n-k)!}{n!}\, b_k\, \partial_x^k. \tag{2}$$

The coefficients of $r = p \boxplus_n q$, $r(x) = \sum_k c_k x^{n-k}$, satisfy

$$c_k = \sum_{i+j=k} \frac{(n-i)!\,(n-j)!}{n!\,(n-k)!}\, a_i\, b_j. \tag{3}$$

Two fundamental properties:

Theorem 2.2 ([4])**.** *If $p, q \in \mathcal{P}_n^{\mathbb{R}}$, then $p \boxplus_n q \in \mathcal{P}_n^{\mathbb{R}}$.*

Lemma 2.3 (Variance additivity)**.** $\sigma^2(p \boxplus_n q) = \sigma^2(p) + \sigma^2(q)$.

Proof. From (3), $c_1 = a_1 + b_1$ and $c_2 = a_2 + \frac{n-1}{n} a_1 b_1 + b_2$. Substituting into Lemma 2.1 and expanding $(a_1 + b_1)^2$, the cross-terms $\frac{2(n-1)a_1 b_1}{n^2}$ and $-\frac{2(n-1)a_1 b_1}{n^2}$ cancel. □

2.3. Scores and Fisher information.

Definition 2.4. For $p \in \mathcal{P}_n^{\mathbb{R}}$ with distinct roots $\lambda_1 < \cdots < \lambda_n$, the *score* at λ_i is $V_i := \sum_{j \neq i} (\lambda_i - \lambda_j)^{-1}$, the *Fisher information* is $\Phi_n(p) := \sum_i V_i^2$, and the *score-gradient energy* is $\mathcal{S}(p) := \sum_{i<j} (V_i - V_j)^2/(\lambda_i - \lambda_j)^2$.

Lemma 2.5. $V_i = p''(\lambda_i)/(2\,p'(\lambda_i))$.

Proof. Since $p'(\lambda_i) = \prod_{j \neq i}(\lambda_i - \lambda_j)$, differentiating once more gives $p''(\lambda_i) = 2 \sum_{k \neq i} \prod_{j \neq i,\, j \neq k} (\lambda_i - \lambda_j) = 2\,p'(\lambda_i)\,V_i$. □

Lemma 2.6 (Score identities)**.**
(i) $\sum_i V_i = 0$.
(ii) $\sum_i \lambda_i V_i = \binom{n}{2}$.
(iii) $\sum_i (\lambda_i - \bar{\lambda}) V_i = \binom{n}{2}$.
(iv) $\Phi_n(p) = \sum_{i<j} \dfrac{V_i - V_j}{\lambda_i - \lambda_j}$.

Proof. (i): $\sum_i V_i = \sum_{i \neq j} (\lambda_i - \lambda_j)^{-1} = 0$ by antisymmetry.
(ii): $\sum_i \lambda_i V_i = \sum_{i \neq j} \lambda_i/(\lambda_i - \lambda_j) = \sum_{i<j} [\lambda_i/(\lambda_i - \lambda_j) + \lambda_j/(\lambda_j - \lambda_i)] = \sum_{i<j} 1 = \binom{n}{2}$.
(iii): Immediate from (ii) and (i).
(iv): $\sum_i V_i^2 = \sum_{i \neq j} V_i/(\lambda_i - \lambda_j) = \sum_{i<j} (V_i - V_j)/(\lambda_i - \lambda_j)$. □

Lemma 2.7 (Fisher–variance inequality)**.** $\Phi_n(p) \cdot \sigma^2(p) \geq n(n-1)^2/4$.

Proof. By Cauchy–Schwarz applied to Lemma 2.6(iii): $\binom{n}{2}^2 \leq \left(\sum (\lambda_i - \bar{\lambda})^2\right)\left(\sum V_i^2\right) = n\,\sigma^2 \cdot \Phi_n$. □

3. The Score-Gradient Inequality

Theorem 3.1 (Score-Gradient Inequality)**.** *For $p \in \mathcal{P}_n^{\mathbb{R}}$ of degree $n \geq 2$ with distinct roots,*

$$\mathcal{S}(p)\,\sigma^2(p) \;\geq\; \frac{n-1}{2}\,\Phi_n(p), \tag{4}$$

with equality if and only if $V_i = c(\lambda_i - \bar{\lambda})$ for some constant c.

Proof. Write $T = n\,\sigma^2$, $U = \Phi_n$, $S = \mathcal{S}$. The claim is $S\,T \geq \frac{n(n-1)}{2}\,U$.

Step 1. By Lemma 2.6(iii) and Cauchy–Schwarz,

$$\frac{n^2(n-1)^2}{4} \;\leq\; T\,U. \tag{5}$$

Step 2. By Lemma 2.6(iv) and Cauchy–Schwarz,

$$U^2 \;\leq\; S \cdot \binom{n}{2}. \tag{6}$$

Step 3. Combining: $S\,T \geq \frac{2\,U^2}{n(n-1)} \cdot T = \frac{2\,U}{n(n-1)} \cdot T\,U \geq \frac{2\,U}{n(n-1)} \cdot \frac{n^2(n-1)^2}{4} = \frac{n(n-1)}{2}\,U$.

Equality. Equality in (5) holds iff $V_i = c(\lambda_i - \bar{\lambda})$. Equality in (6) holds iff $(V_i - V_j)/(\lambda_i - \lambda_j)$ is constant. These conditions are equivalent: if $V_i = c(\lambda_i - \bar{\lambda})$, then $(V_i - V_j)/(\lambda_i - \lambda_j) = c$ for all $i < j$. □

Remark 3.2. The equality condition $V_i = c(\lambda_i - \bar{\lambda})$ characterises, up to affine transformation, the zeros of the Hermite polynomial H_n: evaluating the ODE $H_n'' - 2xH_n' + 2nH_n = 0$ at a zero x_k yields $V_k = x_k$. For $n = 2$ this holds for all distinct root configurations.

4. The Hermite flow

4.1. The Gaussian semigroup.

Definition 4.1. For $t > 0$, the *finite Gaussian* $g_t \in \mathcal{P}_n^{\mathbb{R}}$ is the polynomial whose roots are at positions $\sqrt{t}\,\xi_1 < \cdots < \sqrt{t}\,\xi_n$, where $\xi_1 < \cdots < \xi_n$ are the zeros of the probabilist Hermite polynomial He_n. Thus $\sigma^2(g_t) = t$. The semigroup property $g_s \boxplus_n g_t = g_{s+t}$ holds, and $g_0 := x^n$.

Fix $p, q \in \mathcal{P}_n^{\mathbb{R}}$ with $a := \sigma^2(p) > 0$ and $b := \sigma^2(q) > 0$.

Definition 4.2 (Hermite flow)**.** Define $p_t := p \boxplus_n g_{bt}$ for $t \geq 0$. Then:
(a) $p_0 = p$;
(b) $\sigma^2(p_t) = a + bt$ (by variance additivity, Lemma 2.3);
(c) $p_t \in \mathcal{P}_n^{\mathbb{R}}$ for all $t \geq 0$ (by Theorem 2.2, since $g_{bt} \in \mathcal{P}_n^{\mathbb{R}}$).

Remark 4.3. In an earlier version of this paper, the flow was defined as $p_t = p \boxplus_n q_t$ where q_t is the fractional power of q in the K-transform ($K_{q_t} = K_q^t$); this gives $q_0 = x^n$, $q_1 = q$, and $\sigma^2(q_t) = tb$. However, q_t need not lie in $\mathcal{P}_n^{\mathbb{R}}$ for non-integer t, so p_t is not guaranteed to be real-rooted along the flow, and the root ODE (7) can fail. The Gaussian semigroup avoids this: $g_{bt} \in \mathcal{P}_n^{\mathbb{R}}$ for all t, the root ODE holds exactly, and all dissipation identities follow. The cost is that the flow reaches $p \boxplus_n g_b$ at $t = 1$, not $p \boxplus_n q$. To bridge this gap, we introduce the Score–Jacobian relation and Jacobian contraction in Section 5.

4.2. Perturbation analysis.

Since $g_{bt} \in \mathcal{P}_n^{\mathbb{R}}$ and $p_t \in \mathcal{P}_n^{\mathbb{R}}$ has simple roots for $t = 0$, a Lyapunov argument establishes simple roots for all $t \geq 0$; we state the key steps.

Lemma 4.4. *The Hermite flow satisfies the backwards heat equation*

$$\partial_t p_t(x) = -\frac{b}{2(n-1)}\, p_t''(x).$$

Proof. The operator $T_{g_{bt}}$ acts via (2) with coefficients of g_{bt}. By the semigroup property $p_{t+h} = p_t \boxplus_n g_{bh}$, and the K-transform of g_{bh} equals $\exp(-bh\, z^2/(2(n-1))) \bmod z^{n+1}$, so the coefficient ODE $\kappa_k' = -b/(2(n-1)) \cdot \kappa_{k-2}$ translates to the stated PDE. □

Lemma 4.5 (Root ODE)**.** *Let $\lambda_i(t)$ denote the roots of p_t. Then*

$$\dot{\lambda}_i(t) = \frac{b}{n-1}\, V_i(t) = \frac{b}{n-1} \sum_{j \neq i} \frac{1}{\lambda_i(t) - \lambda_j(t)}. \tag{7}$$

Proof. Differentiating $p_t(\lambda_i(t)) = 0$ in t: $\partial_t p_t(\lambda_i) + p_t'(\lambda_i)\dot{\lambda}_i = 0$. By Lemma 4.4, $\partial_t p_t(\lambda_i) = -\frac{b}{2(n-1)}\, p_t''(\lambda_i)$, so $\dot{\lambda}_i = \frac{b}{2(n-1)} \frac{p_t''(\lambda_i)}{p_t'(\lambda_i)} = \frac{b}{n-1}\, V_i$ by Lemma 2.5. □

Lemma 4.6 (Simple roots)**.** *For all $t \geq 0$, p_t has n simple real roots.*

Proof. Real-rootedness holds by Theorem 2.2. For simplicity, define the log-Vandermonde $W(t) := \sum_{i<j} \log(\lambda_j(t) - \lambda_i(t))$. By Lemma 4.5,

$$\dot{W}(t) = \sum_{i<j} \frac{\dot{\lambda}_j - \dot{\lambda}_i}{\lambda_j - \lambda_i} = \frac{b}{n-1} \sum_{i<j} \frac{V_j - V_i}{\lambda_j - \lambda_i} = \frac{b}{n-1}\, \Phi_n(p_t) \geq 0,$$

using Lemma 2.6(iv). Hence $W(t) \geq W(0)$, i.e. $\prod_{i<j}(\lambda_j(t) - \lambda_i(t)) \geq \prod_{i<j}(\lambda_j(0) - \lambda_i(0)) =: D_0 > 0$. Since $\sigma^2(p_t) = a + bt$ is bounded on compact intervals, each pairwise gap has a uniform positive lower bound, so the roots remain simple. □

4.3. Dissipation and the integral identity.

Lemma 4.7 (Dissipation)**.** $\dfrac{d}{dt}\Phi_n(p_t) = -\dfrac{2b}{n-1}\, \mathcal{S}(p_t).$

Proof. From Lemma 4.5, write $\varepsilon := bh/(n-1)$ for a small increment h. The perturbed scores satisfy

$$V_i^{(h)} = V_i - \varepsilon \sum_{j \neq i} \frac{V_i - V_j}{(\lambda_i - \lambda_j)^2} + O(h^2).$$

Squaring and summing, then pairing (i,j) with (j,i):

$$\Phi_n(p_{t+h}) = \Phi_n(p_t) - 2\varepsilon \sum_{i<j} \frac{(V_i - V_j)^2}{(\lambda_i - \lambda_j)^2} + O(h^2) = \Phi_n(p_t) - \frac{2bh}{n-1}\,\mathcal{S}(p_t) + O(h^2).$$

Taking $h \to 0$ yields the result (the error terms are uniform by Lemma 4.6). □

Corollary 4.8 (Integral identity)**.**

$$\frac{1}{\Phi_n(p \boxplus_n g_b)} - \frac{1}{\Phi_n(p)} = \frac{2b}{n-1} \int_0^1 \frac{\mathcal{S}(p_t)}{\Phi_n(p_t)^2}\, dt.$$

Proof. Set $f(t) := 1/\Phi_n(p_t)$. By the chain rule and Lemma 4.7, $f'(t) = \frac{2b}{n-1} \cdot \frac{\mathcal{S}(p_t)}{\Phi_n(p_t)^2} \geq 0$. Now integrate from 0 to 1. □

4.4. **Gaussian-input Stam inequality.**

Theorem 4.9 (Gaussian-input Stam)**.** *For all $p \in \mathcal{P}_n^{\mathbb{R}}$ with $\sigma^2(p) = a > 0$ and $t > 0$:*

$$\frac{1}{\Phi_n(p \boxplus_n g_t)} \;\geq\; \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(g_t)}. \tag{8}$$

Proof. Step 1 (Differential inequality). The SGI (Theorem 3.1) applied to $p_s = p \boxplus_n g_{ts}$ gives $\mathcal{S}(p_s) \geq (n-1)\Phi_n(p_s)/(2\sigma^2(p_s))$. Substituting into Lemma 4.7 (with the Gaussian variance parameter t):

$$\frac{d}{ds}\Phi_n(p_s) \;\leq\; -\frac{t}{a+ts}\,\Phi_n(p_s).$$

Integrating $(\log \Phi_n)' \leq -t/(a+ts)$ from 0 to s:

$$\frac{1}{\Phi_n(p_s)} \;\geq\; \frac{a+ts}{a\,\Phi_n(p)}. \tag{9}$$

Step 2 (Integration). Substituting (9) into Corollary 4.8 with $b = t$:

$$\frac{1}{\Phi_n(p \boxplus_n g_t)} - \frac{1}{\Phi_n(p)} \;\geq\; \int_0^1 \frac{t}{(a+ts)\,\Phi_n(p_s)}\, ds \;\geq\; \int_0^1 \frac{t}{a\,\Phi_n(p)}\, ds \;=\; \frac{t}{a\,\Phi_n(p)}.$$

Wait—this gives $1/\Phi_n(p \boxplus_n g_t) \geq (a+t)/(a\,\Phi_n(p))$, which is just (9) at $s = 1$. To get (8), we use the lower bound on the dissipation rate directly.

From Corollary 4.8 and the SGI:

$$f'(s) = \frac{2t}{n-1} \cdot \frac{\mathcal{S}(p_s)}{\Phi_n(p_s)^2} \;\geq\; \frac{t}{\sigma^2(p_s)\,\Phi_n(p_s)} \;\geq\; \frac{t}{\sigma^2(p_s)} \cdot \frac{a+ts}{a\,\Phi_n(p)}.$$

But we can also bound $f'(s)$ from below without using (9): the Cauchy–Schwarz inequality in the L-inner product gives $\mathcal{S}/\Phi_n^2 \geq 1/\binom{n}{2}$ (as in the SGI proof, Step 2), hence

$$f'(s) \;\geq\; \frac{2t}{n-1} \cdot \frac{1}{\binom{n}{2}} = \frac{4t}{n(n-1)^2}.$$

Integrating from 0 to 1:

$$\frac{1}{\Phi_n(p \boxplus_n g_t)} - \frac{1}{\Phi_n(p)} \;\geq\; \frac{4t}{n(n-1)^2} = \frac{1}{\Phi_n(g_t)},$$

since $\Phi_n(g_t) = n(n-1)^2/(4t)$ (the scores of the finite Gaussian satisfy $V_i = \lambda_i/(2t^{1/2} \cdot t^{-1/2}) = \lambda_i/2$... more precisely, by the Hermite ODE $\mathrm{He}_n''(x_k) = x_k\,\mathrm{He}_n'(x_k)$ and scaling, $\Phi_n(g_t) \cdot \sigma^2(g_t) = n(n-1)^2/4$, i.e. equality in the Fisher–variance inequality). □

5. The root map and Jacobian contraction

The Hermite flow proves the Stam inequality when one input is Gaussian. To handle general q, we need a non-flow ingredient: the *Jacobian contraction* of the root map, which relates the score vectors of p, q, and $p \boxplus_n q$ directly.

5.1. **The root map.**

Definition 5.1. For $p, q \in \mathcal{P}_n^{\mathbb{R}}$ with ordered root vectors $\alpha = (\alpha_1 \geq \cdots \geq \alpha_n)$ and $\beta = (\beta_1 \geq \cdots \geq \beta_n)$, define $\Omega(\alpha, \beta) \in \mathbb{R}^n$ to be the ordered root vector of $p \boxplus_n q$. Write $\gamma := \Omega(\alpha, \beta)$. Let J denote the $n \times 2n$ Jacobian matrix of Ω at (α, β), defined whenever the roots are simple.

5.2. **Score–Jacobian relation.**

Proposition 5.2 (Score–Jacobian relation)**.** *Let $p, q \in \mathcal{P}_n^{\mathbb{R}}$ have simple roots with ordered root vectors α, β and $\gamma = \Omega(\alpha, \beta)$. For any $a, b \geq 0$:*

$$J \begin{pmatrix} a\, V(\alpha) \\ b\, V(\beta) \end{pmatrix} = (a+b)\, V(\gamma), \tag{10}$$

where $V(\alpha) = \big(\sum_{j \neq i} (\alpha_i - \alpha_j)^{-1}\big)_{i=1}^n$.

Proof. For $t \geq 0$, define $p_t := \exp(-\frac{t}{2}\partial_x^2)\, p$ with ordered root vector $\alpha(t)$, and similarly q_t, r_t with $\beta(t)$, $\gamma(t)$. Since $\boxplus_n$ commutes with any differential operator (because the K-transform converts differentiation to multiplication), $r_{(a+b)t} = p_{at} \boxplus_n q_{bt}$, whence $\gamma((a+b)t) = \Omega(\alpha(at), \beta(bt))$. Differentiating at $t = 0$:

$$(a+b)\, \gamma'(0) = J \begin{pmatrix} a\, \alpha'(0) \\ b\, \beta'(0) \end{pmatrix}.$$

By Lemma 2.5, $\alpha_i'(0) = p''(\alpha_i)/(2\, p'(\alpha_i)) = V_i(\alpha)$, and similarly for β and γ. □

5.3. **Jacobian contraction.**

Proposition 5.3 (Jacobian contraction)**.** *Let $\mathcal{V} := \{(u, v) \in \mathbb{R}^n \times \mathbb{R}^n : u^T \mathbf{1} = v^T \mathbf{1} = 0\}$. For all $(u, v) \in \mathcal{V}$:*

$$\|J(u, v)\|^2 \;\leq\; \|u\|^2 + \|v\|^2. \tag{11}$$

Proof. Fix $(u, v) \in \mathcal{V}$ and consider the paths $\alpha(t) = \alpha + tu$, $\beta(t) = \beta + tv$, $\gamma(t) = \Omega(\alpha(t), \beta(t))$. By variance additivity (Lemma 2.3) applied to root vectors:

$$\frac{1}{n} \sum_i \gamma_i(t)^2 - \Big(\frac{1}{n} \sum_i \gamma_i(t)\Big)^2 = \mathrm{Var}(\alpha(t)) + \mathrm{Var}(\beta(t)).$$

Since $(u, v) \in \mathcal{V}$, the means are constant in t. Differentiating twice at $t = 0$:

$$\frac{1}{n} \sum_i [(D_w \gamma_i)^2 + \gamma_i\, D_w^2 \gamma_i] = \frac{1}{n}(\|u\|^2 + \|v\|^2),$$

where $w = (u, v)$, giving the *Hessian identity*

$$\|Jw\|^2 = \|u\|^2 + \|v\|^2 - \sum_i \gamma_i\, w^T H^{(i)} w, \tag{12}$$

with $H^{(i)} := \mathrm{Hess}\, \Omega_i$ at (α, β). It remains to show $\sum_i \gamma_i\, w^T H^{(i)} w \geq 0$.

The polynomial $f(x, a, b) := \mathbb{E}_{\pi \in S_n} \prod_{i=1}^n (x - a_i - b_{\pi(i)})$ is hyperbolic in the direction $e_1 = (1, 0, \ldots, 0)$. By a theorem of Bauschke, Güler, Lewis, and Sendov [1], the partial sums $\sigma_k(a, b) := \sum_{i=1}^k \gamma_i(a, b)$ are convex, so $\mathrm{Hess}\, \sigma_k \succeq 0$. Since $\gamma_1 \geq \cdots \geq \gamma_n$, $\sum_i \gamma_i\, H^{(i)}$ is a positive linear combination of $\mathrm{Hess}\, \sigma_k$, hence $\sum_i \gamma_i\, w^T H^{(i)} w \geq 0$. □

6. Proof of the main theorem

Theorem 6.1. *Inequality* (1) *holds for every* $n \geq 2$.

Proof. We may assume p and q have simple roots (the extension to repeated roots is in Remark 6.3). Write α, β, γ for their ordered root vectors.

Step 1 (Contraction applied to scores). By Lemma 2.6(i), $(V(\alpha), V(\beta)) \in \mathcal{V}$. Apply Proposition 5.3 with $u = a\,V(\alpha)$, $v = b\,V(\beta)$ ($a, b > 0$ to be chosen):

$$\|J(a\,V(\alpha),\, b\,V(\beta))\|^2 \leq a^2\,\Phi_n(p) + b^2\,\Phi_n(q).$$

By Proposition 5.2, the left-hand side equals $(a+b)^2\,\Phi_n(p \boxplus_n q)$. Hence

$$(a+b)^2\,\Phi_n(p \boxplus_n q) \;\leq\; a^2\,\Phi_n(p) + b^2\,\Phi_n(q). \tag{13}$$

Step 2 (Blachman's choice). Set $a = 1/\Phi_n(p)$ and $b = 1/\Phi_n(q)$. Then (13) becomes

$$\Big(\frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)}\Big)^2 \Phi_n(p \boxplus_n q) \;\leq\; \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)}.$$

Dividing both sides by $\big(1/\Phi_n(p) + 1/\Phi_n(q)\big)\,\Phi_n(p \boxplus_n q) > 0$ yields (1). □

Remark 6.2. The proof above replaces the case-split argument from the original version of this paper, in which the forward bound $1/\Phi_n(p \boxplus_n q) \geq (a+b)/(a\,\Phi_n(p))$ and its reverse were combined. Those bounds followed from flowing p along q using fractional K-transform powers; however, the fractional power q_t is not generally real-rooted (see Remark 4.3), so the flow does not preserve $\mathcal{P}_n^{\mathbb{R}}$ and the root ODE used in the Lyapunov argument is invalid. The Jacobian contraction (Proposition 5.3) provides the missing bridge.

Remark 6.3. With the convention $1/\Phi_n := 0$ for repeated roots, (1) extends to the boundary of $\mathcal{P}_n^{\mathbb{R}}$. When both p and q have repeated roots, both sides vanish. When exactly one, say p, has a repeated root, approximate p by simple-root polynomials $p_\varepsilon \to p$; the inequality for p_ε, q gives $1/\Phi_n(p_\varepsilon \boxplus_n q) \geq 1/\Phi_n(q)$, and continuity of the convolution yields the result in the limit. The Nuij operator $T_\varepsilon := (1 - \varepsilon\, d/dx)^n$ provides such approximations explicitly: $T_\varepsilon p$ is monic, real-rooted, and has simple roots for $\varepsilon > 0$; moreover, T_ε commutes with $\boxplus_n$, so $(T_\varepsilon p) \boxplus_n (T_\varepsilon q) = T_\varepsilon^2 (p \boxplus_n q)$.

References

References

[1] H. H. Bauschke, O. Güler, A. S. Lewis, and H. S. Sendov, *Hyperbolic polynomials and convex analysis*, Canad. J. Math. **53** (2001), 470–488.

[2] N. M. Blachman, *The convolution inequality for entropy powers*, IEEE Trans. Inform. Theory **IT-11** (1965), 267–271.

[3] W. Nuij, *A note on hyperbolic polynomials*, Math. Scand. **23** (1968), 69–72.

[4] A. Marcus, D. A. Spielman, and N. Srivastava, *Interlacing families II: Mixed characteristic polynomials and the Kadison–Singer problem*, Ann. of Math. (2) **182** (2015), 327–350.

[5] A. J. Stam, *Some inequalities satisfied by the quantities of information of Fisher and Shannon*, Inform. Control **2** (1959), 101–112.